NARENDRA SANKARAPU

B.Tech - NBKR Institute of Technology and Science

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Senior C++ Embedded Software Engineer (Experience: 5+ years)

Career Objective

To acquire a challenging career in embedded software development where my skills can be utilized in maximizing business growth and delivering high-quality automotive solutions.

Professional Summary

- Senior Software Engineer with **5+ years** of experience in **Embedded Automotive domain** under **Linux and QNX platforms**
- Expertise in **C++, Embedded C**, **AUTOSAR architecture**, and **ROS (Robot Operating System)**
- Experience with **LIDAR, sensor fusion, camera integration** for mining automation and automotive systems
- Proficient in **Vector Cast, GDB debugging**, and real-time embedded systems development
- Strong background in **HMI development, driver development (SPI, I2C)**, and multi-threading
- Experienced in **Agile/Scrum**, proven track record of increasing code coverage (65%→85%, 70%→88%)

Technical Skills

Languages & Platforms: C++, Embedded C, Pro C | QNX, Linux, Windows

Development Tools: Visual Studio, GDB, Vector Cast, Oscilloscope, Logic Analyzer

Protocols & Communication: I2C, CAN, SPI, UART, TCP/IP, Socket Programming

Version Control & CI/CD: GIT, GitHub, GitLab, Bitbucket, Jenkins, jFrog, Azure DevOps

Frameworks & Standards: AUTOSAR, ROS (rviz, rqt), Qt Framework | JIRA, Confluence, DOORS

Core Concepts: OOPs, STL, Multi-Threading, Design Patterns, Templates, Smart Pointers, IPC, Mutex, Semaphore, RTOS

Professional Experience

Current: CATERPILLAR (Payroll: L&T Technology Services) | April 2025 - Present

Position: Senior Software Engineer | **Domain:** Embedded Automotive - Mining Equipment

Previous: Cyient | December 2020 - April 2025 (4+ years)

Position: Software Engineer | **Domain:** Embedded Automotive

PROJECT #1: Integration of ARCAM Application into AUTOSAR (Cyient)

Role: Software Engineer | **Team:** 6 | **Duration:** 2021-2024

Project Details: Integrated augmented reality (AR) camera system with automotive head units (HUs) using AUTOSAR architecture. ARCAM App acts as interface between AR camera and client applications, delivered as library for seamless AUTOSAR integration.

Responsibilities:

- Analyzed business requirements and conducted approach meetings with teams and stakeholders
- Implemented features based on AUTOSAR specifications and MISRA C++ coding standards
- Integrated ARCAM library with AUTOSAR BSW layer, performed code reviews and compliance checks
- Developed AR camera communication modules ensuring proper data flow between camera and applications
- Conducted unit/integration testing using Vector Cast, increased code coverage 65% to 85%
- Fixed JIRA-reported bugs, delivered builds to QA, documented specifications in Confluence

Tech Stack: C++, AUTOSAR, AR Camera, Vector Cast, Linux, QNX, Visual Studio, GIT, JIRA

PROJECT #2: Development of Human Machine Interface (HMI) Applications (Cyient)

Role: Senior Software Engineer | **Team:** 8 | **Duration:** 2022-2025

Project Details: Developed HMI applications (Settings, Multimedia modules) for four-wheeler car head units. In-vehicle infotainment system manages settings, multimedia playback (audio/video), providing intuitive UI for drivers and passengers.

Responsibilities:

- Analyzed code flow and APIs, enhanced features based on customer requirements in Agile sprints
- Developed Settings module (display, audio, connectivity) and Multimedia module (MP3, AAC, MP4, AVI)
- Fixed bugs from code analysis tools (Coverity, Klocwork), involved in release process
- Conducted manual testing, provided QA knowledge transfer, documented specifications in Confluence
- Provided technical support to team members for on-time delivery

Tech Stack: C++, Qt Framework, HMI Development, Linux, QNX, Windows, Visual Studio, JIRA, GIT

PROJECT #3: Command for Underground (CFU) - Hard Mining Applications (CATERPILLAR)

Role: Senior Software Engineer | **Team:** 7 | **Duration:** April 2025 - Present

Project Details: Developed embedded software to collect and process information from underground hard mining operations. System collects real-time data from LIDARs, sensors, cameras on underground mining trucks, visualizes data to operators for safe remote mining operations.

Responsibilities:

- Analyzed user stories and code architecture, participated in design reviews following Agile methodology
- Integrated LIDAR, sensor, camera data streams using embedded C++ with real-time performance focus
- Implemented data processing modules using multi-threading, developed visualization modules
- Performed dev testing using ROS tools (rviz, rqt), analyzed bugs using GDB and ROS diagnostics
- Worked with I2C/SPI drivers for sensor communication, optimized protocols to reduce latency
- Increased code coverage 70% to 88%, delivered builds in Agile sprints, conducted ROS seminars

Tech Stack: C++, ROS (rviz, rqt), LIDAR, Sensor Fusion, Camera Drivers, Linux, Azure DevOps, GitHub, Jenkins, GDB, I2C, SPI, Multi-threading

Other Responsibilities & Achievements

- Mentored junior developers on embedded C++ best practices, design patterns, and automotive coding standards
- Conducted code reviews ensuring code quality, maintainability, and adherence to guidelines
- Improved system performance through code optimization, reducing CPU usage by 15%
- Conducted technical seminars on ROS concepts, AUTOSAR architecture, and embedded best practices

Educational Qualifications

B.Tech | NBKR Institute of Technology and Science, JNTU-Anantapur | 2018 | CGPA: 6.0/10.0

Personal Information

Languages: English, Telugu, Hindi | **Nationality:** Indian

Declaration

I hereby declare that all the information furnished above is true to the best of my knowledge and belief.

Place: Chennai | Date: May 05, 2026

(NARENDRA SANKARAPU)